Day 16 Problem Set

Geometry Summer Credit | Circle Vocabulary, Equations, and Graphing

Name:	Date:	Period:

Directions

Show center and radius before graphing or writing an equation. Remember that the center signs are opposite and the right side is r^2.

Part A: Vocabulary and Relationships

1.	Define a circle using the words center and distance.

2.	A circle has radius 11. Find the diameter.

3.	A circle has diameter 34. Find the radius.

4.	Name the segment with endpoints on a circle that passes through the center.

5.	Name the line that touches a circle at exactly one point.

6.	Name the line that intersects a circle at two points.

Part B: Read and Write Equations

7.	State the center and radius: $(x - 5)^2 + (y + 3)^2 = 49$.

8.	State the center and radius: $(x + 4)^2 + (y - 6)^2 = 81$.

9.	State the center and radius: $x^2 + y^2 = 36$.

10.	Write the equation of a circle with center (2, -7) and radius 5.

11.	Write the equation of a circle with center (-3, -1) and radius 8.

12.	Write the equation of a circle centered at the origin with diameter 18.

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Continued | Graphing, Points, and Algebra Preview

13.	For $(x - 1)^2 + (y - 4)^2 = 9$, list the center, radius, and four anchor points.
14.	For $(x + 2)^2 + (y + 3)^2 = 25$, list the center, radius, and four anchor points.
15.	Write an equation for a circle centered at (3, 2) that passes through (3, 8).
16.	Write an equation for a circle centered at (-1, 5) that passes through (4, 5).
17.	Determine whether (7, -1) lies on $(x - 3)^2 + (y + 1)^2 = 16$. Show substitution.
18.	Determine whether (1, 6) lies on $(x + 2)^2 + (y - 2)^2 = 25$. Show substitution.
19.	Error analysis: A student reads the center of $(x + 5)^2 + (y - 2)^2 = 36$ as (5, -2). Explain and correct the error.
20.	Error analysis: A student says the radius of $(x - 4)^2 + y^2 = 64$ is 64. Explain and correct the error.
21.	Preview: Rewrite $x^2 - 8x + y^2 + 2y = 8$ in standard form by completing the square.
22.	Challenge: A circle has center (2, -1) and passes through (8, 7). Find its radius and write its equation.